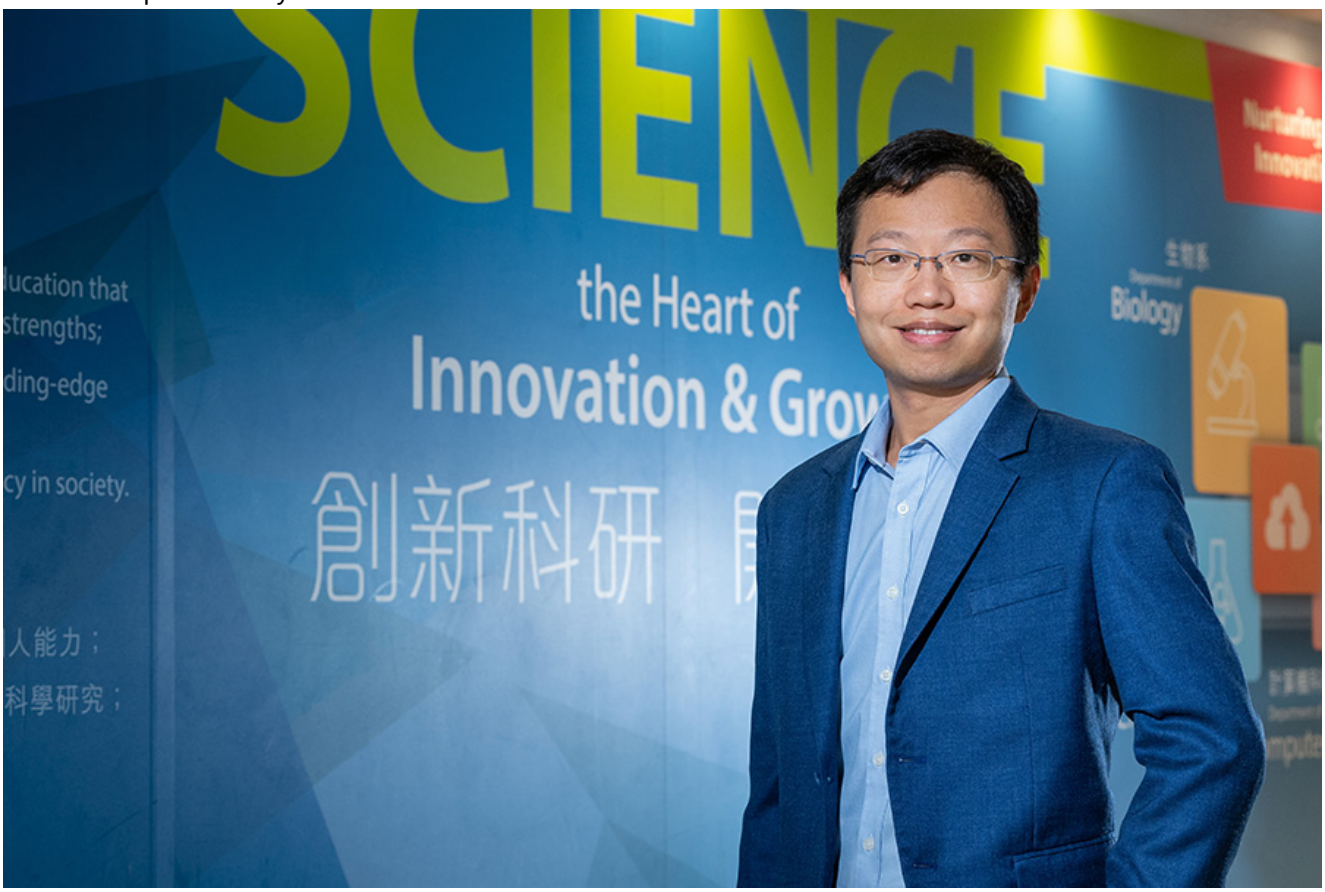

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About

Dr. Liu is currently an Assistant Professor in the Department of Computer Science at Hong Kong Baptist University and the Associate Director of the Health Informatics Center. He received his B.Eng. and M.Eng. degrees in Automation from National University of Defense Technology in 2004 and 2007, respectively. He received the Ph.D. degree in Computing from The Hong Kong Polytechnic University in 2011. During Feb.-Aug. 2010, he was a Visiting Scholar in the Robotics Institute at Carnegie Mellon University. Between 2011 and 2012, he was a Postdoctoral Research Associate in the Department of Statistics at Yale University. Dr. Liu's research interests include artificial intelligence, machine learning, and applied mathematics, as well as their applications in high-dimensional/heterogeneous data analytics, computational epidemiology, and infectious disease modeling. He has published more than 100 peer-reviewed papers in reputable venues, including top-tier journals such as The Lancet Discovery Science, AIJ, T-NNLS, T-Cyber, T-IP, T-AC, T-AMD, T-IST, PR, and NeuroImage, as well as top-tier conferences such as AAAI, IJCAI, SIGIR, ACMMM, WWW, and CIKM.

Research Interests

- Artificial Intelligence, Machine Learning, Pattern Recognition
- Dimensionality Reduction, Subspace Learning
- Multi-way/Multi-view/Multi-label/Multi-task Learning
- Complex Dynamical Systems Modeling
- Computational Epidemiology, Infectious Disease Modeling

Selected Publications

- Mutong Liu, **Yang Liu**, and Jiming Liu, Machine learning for infectious disease risk prediction: A survey, *ACM Computing Survey (**CSUR**)*, 2025. Accepted for Publication.
- Valentina Franzoni, Giulio Biondi, **Yang Liu**, and Alfredo Milani, Evolving meta-correlation classes for binary similarity, *Pattern Recognition (**PR**)*, 157:110871, 2025.
- Tiantian He, **Yang Liu**, Yew-Soon Ong, Xiaohu Wu, Xin Luo, Polarized message-passing in graph neural networks, *Artificial Intelligence (**AIJ**)*, 331, 104129, 2024.
- Jinfu Ren, **Yang Liu**, and Jiming Liu, Consecutive one-week model predictions of land surface temperature stay on track for a decade with chaotic behavior tracking, *Communications Earth & Environment, Nature Portfolio*, 5:627, 2024.
- Jinfu Ren, **Yang Liu**, and Jiming Liu, Commonality and individuality based subspace learning, *IEEE Transactions on Cybernetics (**TCYB**)*, 54(3), pp. 1456-1469, 2024.
- Qi Tan, **Yang Liu**, and Jiming Liu, Mutually adaptable learning, *IEEE Transactions on Emerging Topics in Computational Intelligence (**TETCI**)*, 8(1), pp. 240-254, 2024.

- Jinfu Ren, **Yang Liu**, and Jiming Liu, Chaotic behavior learning via information tracking, *Chaos, Solitons & Fractals*, 175(1), 113927, 2023.
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- **Yang Liu**, Zhonglei Gu, Shang Xia, Benyun Shi, Xiao-Nong Zhou, Yong Shi, and Jiming Liu, What are the underlying transmission patterns of COVID-19 outbreak? – An age-specific social contact characterization, *EClinicalMedicine, The LANCET Discovery Science*, 22, 100354, May, 2020.
- Tiantian He*, **Yang Liu***, Tobey H. Ko, Keith C.C. Chan, and Yew-Soon Ong, Contextual Correlation Preserving Multi-View Featured Graph Clustering, *IEEE Transactions on Cybernetics (**TCYB**)*, 50(10), pp. 4318-4331, 2020. (*Joint First Author)
- **Yang Liu**, Zhonglei Gu, Tobey H. Ko, and Jiming Liu, Identifying Key Opinion Leaders in Social Media via Modality-Consistent Harmonized Discriminant Embedding, *IEEE Transactions on Cybernetics (**TCYB**)*, 50(2), pp.717–728, 2020.